

Machine Learning of Forced Convection Heat Transfer

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Abstract. Prediction of the convective heat transfer coefficient (or Nusselt number) and heat transfer rates in air-cooled systems are crucial for optimizing the performance of thermal systems. Higher rates of heat transfer indicate efficient thermal convection. Accurate predictions of heat transfer rates help determine fluid flow rates, appropriate materials for construction, and other operating conditions for optimal performance of thermal systems (e.g. heat exchangers). This enables the design of energy-efficient systems with improved overall efficiency, which can manage heat effectively at reduced energy consumption rates and operational costs. In addition, the comparison of predicted and actual heat transfer rates enables quantitative evaluation of exergy losses, fouling, or other adverse operational issues. Machine learning (ML) techniques can be employed to assess internal forced convection through ducts by leveraging their ability to analyze complex flow patterns, model non-linear relationships between governing parameters, and predict system behaviour from experimental or simulated data. The present study explores the use of different ML models to predict the Nusselt number and heat transfer rates in two generic configurations of heat exchangers. For the Nusselt number prediction of the first setup, the results indicate that linear regression performs best, achieving an R^2 value of 0.996, an RMSE of 0.798, and an MAE of 0.662. For cold fluid outlet temperature (T_2) prediction of the second configuration, whose performance yielded a different but similar type of dataset, linear regression once again demonstrated good performance with an R^2 value of 0.9967, an RMSE of 0.4913, and an MAE of 0.4002. Similar trends were seen in the prediction of heat transfer rate and the hot fluid outlet temperature. Therefore, ML models present a viable alternative to traditional empirical methods. This integration of ML into thermal system analysis offers new insights into optimizing heat transfer processes and improving efficiency.

Keywords: Convection Heat Transfer · Nusselt Number · Machine Learning

1 Introduction

For various engineering applications, heat exchangers (condensers) that utilize pipes of small diameters are gaining prominence to meet the growing demands for compact and energy-efficient heat transfer solutions. The spectrum of applications includes condensers in cooling and air-conditioning systems, horizontal tubular evaporators in water desalination units, power system heaters, and heat pipes, to mention some. In addition, small-scaled heat exchangers play critical roles in cryogenic processes (e.g. air separation and natural gas liquefaction plants). In these systems, the design and performance of heat exchangers directly impact other key components, such as compressors and drivers. These considerations warrant the development of more precise simulation tools that rely on accurate heat transfer coefficient models [1–4].

As detailed in several publications, some modelling techniques for the prediction of heat transfer coefficients during flow condensation of single-component fluids inside pipes have been developed and validated against experimental data. As the underlying mechanisms that govern heat transfer phenomena are complex, most of these models rely on heuristic approaches. Often, these models incorporate several dimensionless groups and adjustable parameters to fit experimental data. More recently, the application of ML techniques has expanded the complexity of these models (with a greater number of dimensionless groups and adjustable parameters). With this increased complexity, the prediction accuracy for large datasets has improved, even those associated with significant noise and high variability [5, 6].

Machine learning (ML) has emerged as a novel and powerful tool for analyzing and predicting forced convection phenomena and offers innovative approaches to overcome the challenges associated with traditional methods (e.g. empirical correlations, analytical models) and computational fluid dynamics (CFD). ML models (e.g. regression algorithms and neural networks) are trained using experimental and/or CFD data to predict local and average heat transfer coefficients in forced convection systems. These models can incorporate parameters such as the Reynolds number, Prandtl number, Nusselt number, flow velocity, duct geometry, and fluid properties, to predict local and/or average heat transfer rates. Machine learning makes use of classifiers (e.g., decision trees and support vector machines) to distinguish between laminar, transitional, and turbulent flow regimes, based on input features like pressure gradients, and velocity profiles. ML-based surrogate models (e.g. Gaussian processes, deep neural networks) are replacing expensive CFD simulations to provide rapid and accurate predictions of flow and thermal characteristics in forced convection systems. Tools like genetic algorithms or Bayesian optimization, combined with ML models, enable the optimization of heat exchanger designs, cooling systems, and duct configurations to maximize heat transfer rates while minimizing pressure drop and energy consumption. Adaptive control systems, powered by ML, can dynamically adjust operating conditions (e.g. flow rate or temperature) to enable real-time monitoring of forced convection systems, predict performance metrics (e.g. heat transfer rates) and detect anomalies (e.g. flow maldistribution or fouling). In recent times, machine learning has been successfully applied in multiphysics coupling, where complex problems (e.g. conjugate heat transfer in systems with solid-fluid interactions, or fluid flow with phase change) involving forced convection have been solved [7–10].

This study focuses on the application of machine learning models for predicting the Nusselt number and heat transfer rates in thermal systems, with suitable working fluids. It involves utilizing various machine learning tools and techniques, including feature selection, model training, and cross-validation, to analyze and predict thermal behaviour accurately. The workflow of the adopted methodology is shown diagrammatically below (in Fig. 1), illustrating the key steps.

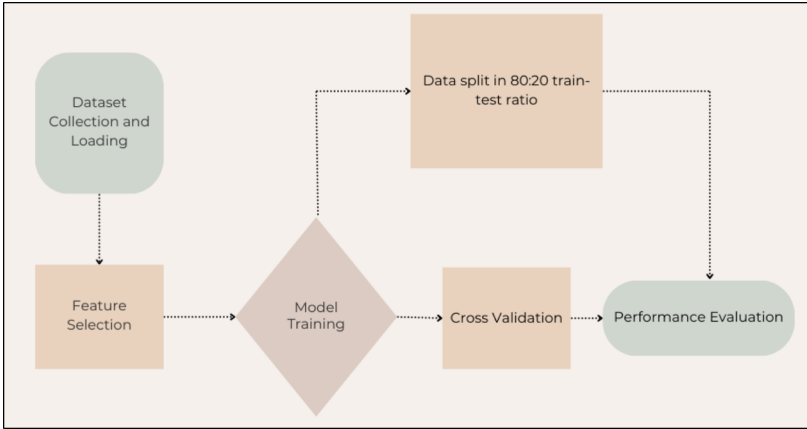


Fig. 1. The work flow of present study

2 Methodology

2.1 Dataset

The present analysis considers two datasets, obtained as outcomes of experimental studies conducted on two generic configurations of heat exchangers. The dataset employed in the first phase of this study consists of 87 experimentally obtained data points designed to evaluate the thermal performance of fluid flow within a tubular heat exchanger. Each data point encapsulates crucial parameters related to the heat transfer rate and the dynamics of fluid flow about the system and ensures a robust foundation for analysis. The dataset is devoid of missing values, thereby ensuring consistency and reliability throughout the study.

In the experimental setup used in this study, the length of the horizontal heat exchanger tube is 1.2 m, and the outer diameter of the heat exchanger tube is $57.5 \times 3.5 \text{ mm}$. The material of the tube (not specified in the dataset) is a high-conductivity material. An external compressor (equipped with a flow control valve) provides a controlled flow of air (working fluid). In addition, a helical wire insert is placed inside the tube (of specified diameter and pitch) to enhance the heat transfer rate. The setup is equipped with calibrated thermocouples (to measure the local temperatures at different sections), flow meter (to measure flow rates) and pressure sensors (for measuring pressure differences). A constant-flux heat source is placed externally, with provisions for monitoring and controlling the power input. The measured values of the inlet and outlet

fluid temperatures, tube wall temperatures, mass flow rate and pressure drop are fed to a data acquisition system installed with the apparatus. The objectives of the study are the determination of local (or average) values of the heat transfer coefficient (or Nusselt number), calculation of the heat transfer rate, and evaluation of the effects of wire diameter and pitch on heat transfer performance and pressure drop.

The salient/key features include geometric and thermal properties such as ‘Wire Diameter’ (mm), ‘Pitch’ (mm), ‘Cold Fluid Inlet Temperature’ (t_1)(°C), and ‘Cold Fluid Outlet Temperature’ (t_2)(°C). In addition, variables related to fluid dynamics and thermophysical properties, which include the ‘Differential Pressure across Orifice Flowmeter’ (ΔP_1) (kPa), the ‘Pipe Wall Temperature’ (t_w)(°C), and the ‘Temperature Difference of Cold Fluid’ (Δt) (°C) are included to provide a comprehensive representation of the underlying physics.

The ‘Cold Fluid Inlet Temperature’ ranges from 26.8 °C to 47.6 °C, with a mean of 34.53 °C and a standard deviation of 4.91 °C, while the ‘Cold Fluid Outlet Temperature’ varies between 78.3 °C and 97.3 °C, averaging 84.28 °C with a standard deviation of 3.93 °C. The ‘Wall Temperature’ spans from 96.7 °C to 100.8 °C, with a mean of 98.76 °C and a standard deviation of 1.18 °C. The ‘Cold Fluid Temperature Difference’ ranges from 34.4 °C to 59.6 °C, averaging 49.74 °C. Regarding flow regimes (laminar or turbulent), the Reynolds Number ranges from 9,303 to 33,609, with a mean of 22,061 and a standard deviation of 5,985, indicating that the dataset predominantly represents turbulent flow.

The ‘Wire Diameter’ and ‘Pitch’ are important parameters in datasets on experimental heat transfer analysis, particularly in systems that involve structured (patterned) surfaces or system configurations promoting enhanced heat transfer (e.g., finned tubes, wire coils, or helical inserts). The wire diameter modifies the flow pattern around the surface; while wires of smaller diameters allow more space for fluid flow, larger diameters increase flow obstruction and turbulence and enhance convective heat transfer rates. Wires of larger diameters (below a threshold value) offer lower thermal resistance and increase the effective surface area available for heat transfer, which can enhance thermal performance. Pitch refers to the spacing between consecutive wires, coils, or fins in a structured heat transfer surface. A smaller pitch (above a threshold value) induces higher turbulence and enhances heat transfer rates, without incurring significant pressure drop. As the pitch affects the density of fins (or wires) per unit length, it alters the heat transfer rate. While a smaller pitch increases the surface area exposed to the fluid and promotes heat transfer rate, a larger pitch reduces the same. A larger value of pitch also ensures more uniform fluid distribution, enabling the reduction of hotspots or stagnation zones.

For the prediction of the Nusselt Number (Nu), which quantifies the convective heat transfer rate, a total of 19 features were selected to capture the influence of geometric parameters of the system, temperature gradients determining the directions of heat flow, and flow dynamics that influence the rate of convective transport. These features include the ‘Wire Diameter’, the ‘Pitch’, the ‘Inlet Air Density’ (ρ_o), the ‘Air Viscosity’ (μ), and the ‘Prandtl Number’ (Pr), among others. To predict the Heat Transfer Rate (Q), a broader range of 22 features were chosen to encompass additional thermal properties such as the ‘Specific Heat Capacity at Constant Pressure’ (C_p), the ‘Thermal Conductivity’ (λ), and the ‘Nusselt Number’ (Nu) itself, given their direct correlations with the

prediction of heat transfer rates. The selection of features was guided by their physical relevance to the prediction tasks, thereby ensuring that the model accurately captures the critical interactions between flow characteristics, thermal gradients, and heat transfer phenomena. For the dataset, one can visit the following link for the Enhanced heat transfer experimental dataset:

<https://www.kaggle.com/datasets/haipotianku/enhanced-heat-transfer-experimental-dataset/data>

The second phase of the study concerns the performance of a plate-fin heat exchanger, which is a compact and efficient device that uses thin metal plates and fins to enhance heat transfer between two (or more) fluid streams. It is characterized by dual attributes of high heat transfer efficiency and lightweight construction. This heat exchanger typically consists of a stack of corrugated fins sandwiched between flat plates, forming a series of counterflow channels. The plates (and fins) are made from materials like aluminium or stainless steel, which offer high thermal conductivity and corrosion resistance. Fluids flow through separate, parallel channels formed by the plates and fins. While thermal energy is transferred through the thin plates, the fins enhance convection by increasing the heat transfer surface area.

The dataset has been obtained from a serialized set of experiments reported in [9]. To assess the performance of the plate-fin heat exchanger, the experiments have been conducted under a steady state. Measurements of temperature and mass flow rates for the ‘hot’ and ‘cold’ fluids provide the required information to compute the heat exchanger effectiveness (ϵ), defined as the ratio of the actual to maximum-possible heat transfer rates. The calculations of this performance parameter are assessed as follows:

$$\epsilon \equiv \frac{\dot{E}_{actual}}{\dot{E}_{max}} = \frac{\Delta T_{minimum fluid}}{\Delta T_{max}} \quad (1)$$

In the metric specified by Eq. (1), the ‘minimum fluid’ is the one that experiences the larger temperature difference in the heat exchanger, and the maximum temperature difference in the heat exchanger is always the difference in inlet temperatures of the ‘hot’ and the ‘cold’ fluids. For the counter-flow heat exchanger, depending on whether the ‘hot’ (or ‘cold’) fluid is the ‘minimum fluid’, the effectiveness (ϵ_h ; ϵ_c) is as follows:

$$\epsilon_h = \frac{\dot{m}_h c_h (T_{hi} - T_{ho})}{\dot{m}_h c_h (T_{hi} - T_{co})} = \frac{(T_{hi} - T_{ho})}{(T_{hi} - T_{co})}; \quad (2)$$

$$\epsilon_c = \frac{\dot{m}_c c_c (T_{ci} - T_{co})}{\dot{m}_c c_c (T_{hi} - T_{co})} = \frac{(T_{hi} - T_{ho})}{(T_{hi} - T_{co})}. \quad (3)$$

In Eqs. (2) and (3), the subscripts ‘i’ and ‘o’ refer to ‘inlet’ and ‘outlet’ respectively. Based on experiments, the dataset has been prepared in [8]; this set, although delimited by the number of experimental trials, comes to use in the assessment of heat exchanger effectiveness for Industrial applications. The experiments were conducted for different values of mass flow rates (5.7 g/s to 14.2 g/s), as well as for different values of hot fluid inlet temperatures (315 K to 365 K). In the context of this dataset, the following variables are defined: ‘Q’ (liters /min) is the mass flow rate, ‘P1’ (kg/cm²) is the pressure recorded at cold fluid inlet, ‘P2’ (kg/cm²) is the pressure at hot fluid inlet, ‘P_{cd}’ (mm of Hg) is the

cold fluid pressure drop, ' P_{hd} ' (mm of Hg) is the pressure drop for hot fluid, ' T_1 ' (K) is the cold fluid inlet temperature, ' T_2 ' (K) is the cold fluid outlet temperature, ' T_3 ' (K) is the hot fluid inlet temperature, and ' T_4 ' (K) is the hot fluid outlet temperature.

2.2 Machine Learning Models

Linear Regression is a foundational statistical method used to establish a relationship between a dependent variable (target or outcome) and one or more independent variables (predictors or features). The primary objective of linear regression is to model this relationship by fitting a linear equation to the observed data, minimizing the differences between the predicted and actual values [11, 12].

In its simplest form, linear regression is represented by the equation:

$$y = \beta_0 + \beta_1 x_1 + \epsilon \quad (4)$$

In Eq. (4), ' y ' is the dependent variable (target), ' β_0 ' is the intercept of the regression line, ' β_1 ' is the coefficient that represents the slope for the independent variable, ' x_1 ' represents the independent variable (feature), and ' ϵ ' is the error term, which accounts for deviations of the observed values from the fitted line. The coefficients β_0 and β_1 are determined by minimizing the sum of squared residuals, which are the differences between the observed and predicted values:

$$\text{Residual Sum of Squares (RSS)} = \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (5)$$

In Eq. (5), y_i is the i^{th} observed value, and $\hat{y}_i$ is the i^{th} predicted value. The linear regression technique can be extended to accommodate multiple independent variables, leading to the following multiple linear regression model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \epsilon \quad (6)$$

In Eq. (6) $x_1, x_2, \dots, x_n$ represents the independent variables, and $\beta_1, \beta_2, \dots, \beta_n$ are their corresponding coefficients. This method is widely employed due to its simplicity, interpretability, and computational efficiency, especially when the relationship between the target and predictors is approximately linear. Moreover, its ability to provide insights into the contribution of individual predictors makes it a valuable tool in both exploratory data analysis and predictive modelling.

In addition to linear regression, the present study incorporates other machine learning techniques to capture complex, non-linear relationships and improve predictive performance. Advanced algorithms such as decision trees, random forests, and support vector machines (SVMs) complement the simplicity of linear regression by addressing its limitations and providing robust solutions for diverse modelling challenges. These methods collectively enhance the study's ability to identify and model intricate patterns in the dataset effectively. Random Forest is an ensemble learning technique that combines the predictions of multiple decision trees to improve predictive accuracy and control overfitting. It operates by constructing a multitude of decision trees during training and aggregating their outputs (either by averaging for regression or voting for classification). The

method leverages the principle of “wisdom of the crowd,” where the collective decision of multiple models outperforms individual models.

Mathematically, for a regression problem, the prediction $\hat{y}$ from a Random Forest (comprising T numbers of decision trees) can be expressed as follows:

$$\hat{y} = \frac{1}{T} \sum_{i=1}^T \hat{y}^{(i)} \quad (7)$$

In Eq. (7) $\hat{y}^{(i)}$ is the prediction from the i^{th} tree. Each decision tree (in the Random Forest) is built using a random subset of features and a random bootstrap sample of the training data. This randomness introduces diversity among the trees and reduces the risk of overfitting, enhancing the model’s generalization ability. The algorithm employs techniques such as:

- **Bootstrap Aggregating (Bagging):** Random subsets of the training data are sampled with replacement to train each tree, ensuring variability in the model inputs.
- **Random Feature Selection:** At each split in a tree, only a random subset of features is considered, reducing correlation among trees and improving robustness.

Random Forest inherently provides feature importance metrics, which enables the identification of influential predictors in the dataset. The importance of a feature is typically assessed based on the mean decrease in impurity or its contribution to prediction accuracy when permuted.

The key advantages of Random Forest include:

- **Non-linear Modeling:** It can capture complex relationships between input features and the target variable.
- **Robustness to Overfitting:** The aggregation of multiple trees mitigates the tendency of overfitting of individual trees.
- **Scalability:** It is efficient for large datasets and high-dimensional feature spaces.

Gradient Boosting Machine (GBM) is another powerful ensemble learning technique that builds predictive models by sequentially combining weak learners, typically decision trees, to minimize errors and improve accuracy. Unlike Random Forest, which trains multiple trees independently in parallel, GBM builds trees in a stage-wise manner, where each new tree corrects the residual errors of the previous trees. This iterative process enables GBM to refine predictions progressively and capture complex, non-linear relationships within the data. Mathematically, at each iteration, a new tree is trained to fit the negative gradient of the loss function, reducing the overall error. GBM employs techniques such as learning rate tuning, regularization, and early stopping to prevent overfitting and enhance model generalization. Its ability to optimize predictive performance while maintaining interpretability makes GBM a widely used approach in regression and classification tasks.

In the thermal analysis presented here, the aforementioned machine learning models have been employed to predict the Nusselt number (Nu) and heat transfer coefficients, all of which are crucial parameters for assessing fluid flow and heat transfer rates in the chosen configuration [10].

2.3 Performance Parameters

To evaluate the efficacy of the predictive models, three performance parameters were used in this study. They are the Coefficient of Determination (R^2), the Root Mean Squared Error (RMSE), and the Mean Absolute Error (MAE).

The R^2 value indicates the proportion of variance in the dependent variable that is explained by the model. Values closer to ‘1’ signify higher accuracy, and demonstrate the model’s ability to explain variability in the data effectively.

The RMSE measures the standard deviation of residuals, which are the differences between the predicted and actual values. This metric provides insight into the overall magnitude of prediction errors, with lower values of RMSE reflecting better model performance.

The MAE quantifies the average absolute difference between the predicted and actual values. It is an interpretable metric that reports errors in the same units as the target variable, making it easy to understand.

Considered together, these metrics offer a comprehensive assessment of model performance. They balance predictive accuracy and error quantification, allowing for robust evaluation across diverse data distributions (Fig. 2).

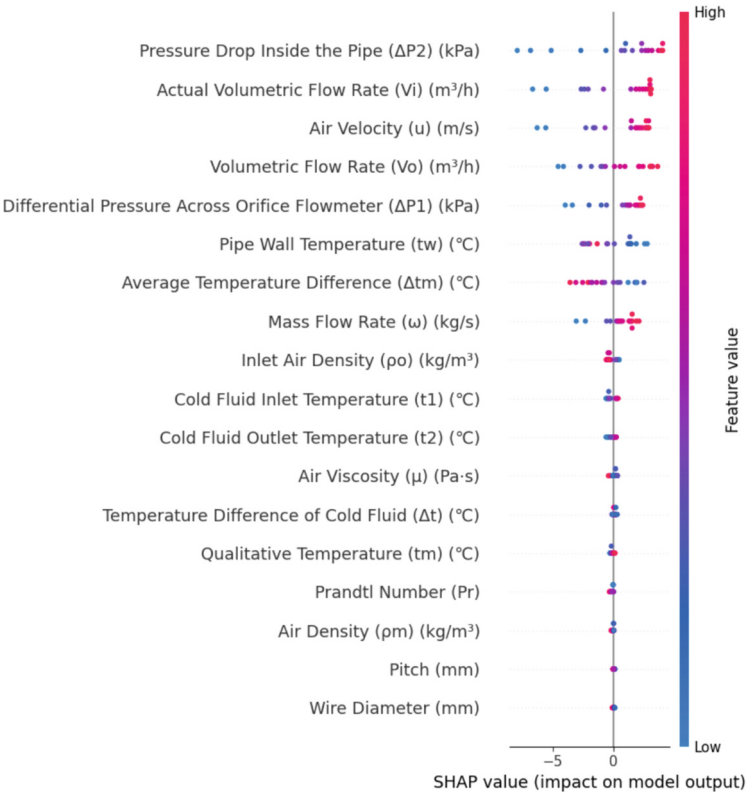


Fig. 2. SHAP Summary Plot for Random Forest Model, depicting feature importance and contribution to predictions for Nusselt Number.

3 Results and Discussion

For the dataset in the first phase of the study, the performance of the models was evaluated using an 80–20 training split and 5-fold cross-validation. The results were assessed based on key regression metrics, including the Coefficient of Determination (R^2), the Root Mean Squared Error (RMSE), and the Mean Absolute Error (MAE). Separate evaluations were conducted to predict the Nusselt Number (Nu) and the Heat Transfer Rate (Q). Based on all the models, the linear factors seemed to overpower the nonlinearity existing in the relationship between the features. To observe the feature importance (see Fig. 2 above), the SHAP summary is provided for the Random Forest model that shows how the most influential features are the pressure drop inside the pipe (ΔP_2), actual volumetric flow rate (V_i), and air velocity (u) in the prediction of Nusselt number.

3.1 Results on 80–20 Training Split

The results for the prediction of the Nusselt Number (Nu) using the 80–20 training split are summarized in Table 1 below. The Linear Regression (LR) has demonstrated the best performance among all the models, achieving an R^2 value of 0.996, an RMSE of 0.798, and an MAE of 0.662. These metrics indicate that Linear Regression has provided highly accurate predictions with minimal error.

The Random Forest (RF) and Gradient Boosting Machines (GBM) also deliver good performance, although their RMSE and MAE were slightly higher as compared to Linear Regression. On the other hand, the Decision Tree (DT) and the Support Vector Regression (SVR) exhibit relatively weaker performance. Among these, the SVR exhibits the least accuracy, with an R^2 value of 0.469, indicating significant room for improvements in its predictions.

Table 1. Results for Nusselt Number (Nu) Prediction (80–20 Split)

Model	R^2	RMSE	MAE
Linear Regression	0.996	0.798	0.662
Decision Tree	0.796	40.716	5.637
Random Forest	0.935	12.939	3.138
Support Vector Machines	0.469	106.180	8.322
Gradient Boosting Machines	0.9295	14.101	3.229

In the context of the prediction of the heat transfer rate (Q), the Linear Regression once again outperformed all other ML models, achieving near-perfect results with an R^2 value of 0.9999, an RMSE of 0.002, and an MAE of 0.035 (refer to Table 2). These metrics underscore the exceptional predictive capability of Linear Regression for this task.

The Random Forest model and Gradient Boosting Machines also produced competitive results with a good R^2 value, although their RMSE and MAE were slightly

higher than those of Linear Regression. In contrast, the Decision Tree and Support Vector Regression exhibited significantly lower accuracy. The SVR, in particular, yielded the lowest R^2 value of 0.131, indicating poor predictive performance for this dataset.

Table 2. Results for Heat Transfer Rate (Q) Prediction (80–20 Split)

Model	R^2	RMSE	MAE
Linear Regression	0.9999	0.002	0.035
Decision Tree	0.857	461.901	16.811
Random Forest	0.925	243.847	10.658
Support Vector Machines	0.131	2811.377	40.223
Gradient Boosting Machines	0.9659	110.2436	8.7924

3.2 Results on 5-Fold Cross-Validation

The robustness of the models was further validated using 5-fold cross-validation. For Nu prediction (Table 3; see below), LR maintained exceptional performance, achieving a perfect mean $R^2 = 1.0$, while RF delivered consistent results with a mean $R^2 = 0.89$. SVR’s performance declined further in this scenario ($R^2 = 0.26$), highlighting its limitations in capturing the relationships in the dataset.

Table 3. Fold Cross-Validation Results for Nu Prediction

Model	Mean R^2	Mean RMSE	Mean MAE
Linear Regression	1.0	1.03	0.76
Decision Tree	0.83	6.68	5.43
Random Forest	0.89	5.64	4.28
Support Vector Machines	0.26	14.53	11.88
Gradient Boosting Machines	0.92	4.79	3.76

For Q prediction (refer to Table 4), Linear Regression again excelled with mean $R^2 = 1.0$ and low errors (RMSE: 0.06, MAE: 0.04), as shown below. Random Forest demonstrated reliable performance with mean $R^2 = 0.93$, while Support Vector Machines showed significant deviations ($R^2 = -0.11$), indicating poor generalization.

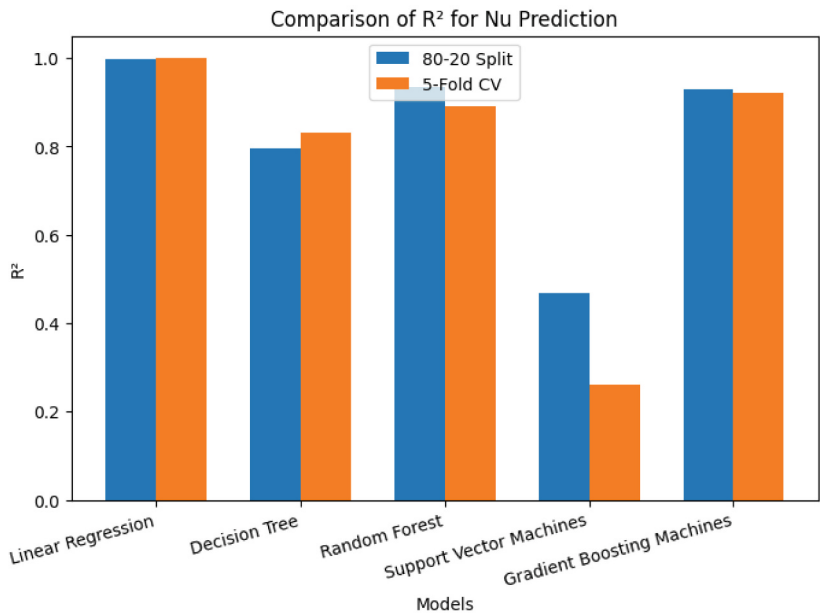
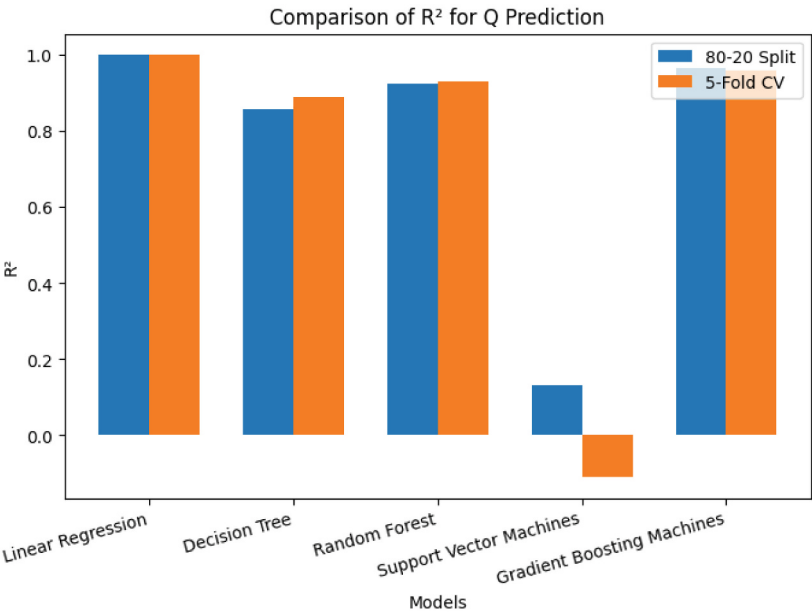


Table 4. Fold Cross-Validation Results for Q Prediction

Model	Mean R ²	Mean RMSE	Mean MAE
Linear Regression	1.0	0.06	0.04
Decision Tree	0.89	21.51	17.35
Random Forest	0.93	16.99	12.52
Support Vector Machines	−0.11	69.83	55.32
Gradient Boosting Machines	0.96	13.55	10.79

Linear Regression (LR) and Random Forest (RF) outperformed Decision Trees (DT) and Support Vector Regression (SVR) in predicting forced convection heat transfer rates, due to their ability to capture the dataset’s underlying relationships effectively. LR achieved near-perfect accuracy ($R^2 \approx 1.0$), indicating a strong linear correlation between features aligning with fundamental heat transfer equations. RF, while slightly less accurate than LR, handled localized non-linearities (e.g., turbulence effects) better through ensemble learning, feature importance ranking, and noise reduction. In contrast, DT suffered from overfitting, leading to high variance, while SVR performed the worst ($R^2 \approx 0.13$ for heat transfer rate prediction), likely due to its reliance on complex kernel transformations, which were unnecessary given the dataset’s strong linear trends. Future work could explore deep learning approaches for more complex heat transfer systems with strong non-linearity, provided a comprehensive dataset is aggregated.

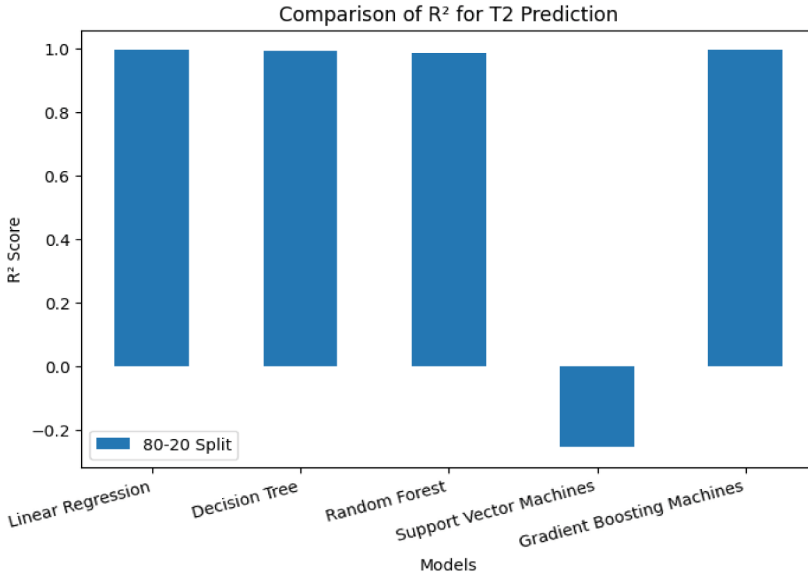


When the above models were applied to the second dataset (assessing the effectiveness of plate-fin heat exchanger), some results were consistent with the previous ones, while others showed some deviations.

Table 5. Results for Cold fluid outlet temperature (T2) (80–20 Split)

Model	R ²	RMSE	MAE
Linear Regression	0.9967	0.4913	0.4002
Decision Tree	0.9937	0.6823	0.5640
Random Forest	0.9875	0.9596	0.8918
Support Vector Machines	−0.2555	9.6203	8.2579
Gradient Boosting Machines	0.9970	0.4732	0.4085

The performance of the models in predicting the cold fluid outlet temperature (T2) followed a similar trend observed in the previous experiments, with Linear Regression (LR) and Gradient Boosting Machines (GBM) achieving the highest accuracy. Table 5 summarizes the results for T2 prediction using the 80–20 training split. GBM demonstrated the best performance, achieving the highest R² value of 0.9970 and the lowest RMSE (0.4732) and MAE (0.4085), indicating superior predictive capability. LR also performed exceptionally well, with an R² of 0.9967, an RMSE of 0.4913, and an MAE of 0.4002, making it a strong competitor to GBM.



The Decision Tree (DT) model showed slightly lower accuracy than LR and GBM, with an R^2 of 0.9937, but its RMSE (0.6823) and MAE (0.5640) were higher, indicating that it introduced more errors in prediction. Meanwhile, Random Forest (RF) also tried to maintain good performance.

In contrast, Support Vector Regression (SVR) exhibited significantly poor performance, with an R^2 value of -0.2555, an RMSE of 9.6203, and an MAE of 8.2579. These results suggest that SVR struggled to generalize well for T2 prediction, likely due to the dataset's inherent linear relationships, which made the complex kernel transformations ineffective.

The noticeable errors observed in the Support Vector Regression (SVR) model align with findings from previous studies, such as those utilizing Artificial Neural Networks (ANN) for plate-fin heat exchanger performance prediction [8]. In [8], despite ANN's capability to capture complex nonlinear relationships, significant errors were reported in certain cases, particularly when the dataset exhibited strong linear correlations. This supports the observation (of this study) that SVR, which relies on kernel transformations to map data into higher-dimensional spaces, struggles to generalize effectively and leads to poor predictive performance.

SVR's high RMSE and MAE values suggest that its kernel-based approach was not well-suited for this dataset, likely due to the dominance of linear dependencies in heat exchanger parameters. Similar trends were noted in the reference study, where ANN models exhibited noticeable deviations, reinforcing the idea that complex models may not always outperform simpler regression-based approaches when dealing with datasets with inherent linearity.

Overall, the findings reaffirm that LR and GBM are the most reliable models for predicting the cold fluid outlet temperature (T2) due to their ability to capture the dominant patterns in the dataset effectively. Future research could further explore hybrid

models or deep learning approaches to enhance predictive accuracy for more complex heat exchanger systems.

The performance evaluation for predicting the hot fluid outlet temperature (T4) using an 80–20 training split reveals notable differences across the models, as seen in Table 6. Linear Regression (LR) outperformed other models, achieving the highest R^2 value of 0.9336 and the lowest RMSE (0.3917) and MAE (0.2907), indicating its superior ability to capture the linear relationships in the dataset.

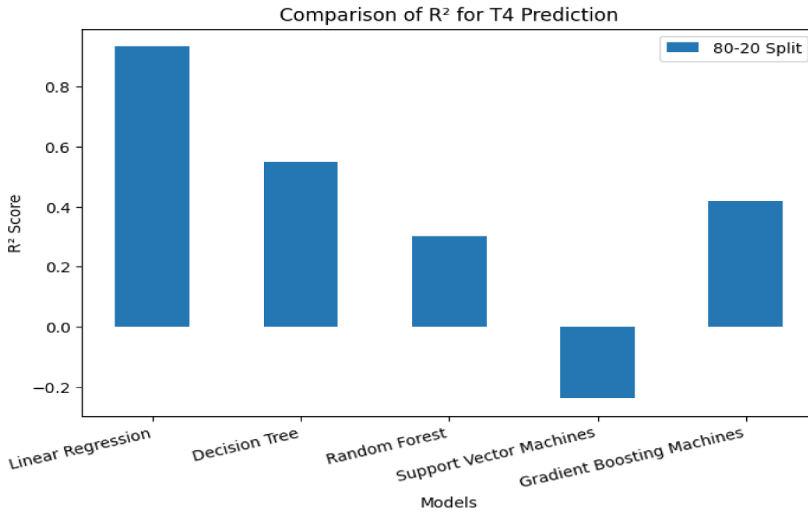
Gradient Boosting Machines (GBM) demonstrated moderate performance, with an R^2 value of 0.4191 and an RMSE of 1.1580, but its MAE (0.8613) was significantly higher than LR. The Decision Tree (DT) model showed limited accuracy, with an R^2 value of 0.5498 and higher RMSE (0.8760) and MAE (1.0195), indicating potential overfitting and reduced generalization.

Random Forest (RF) and Support Vector Machines (SVR) exhibited the weakest performance. RF achieved an R^2 of only 0.3011 and reported higher error metrics (RMSE: 1.2703, MAE: 1.6136), while SVR performed the worst, with a negative R^2 value (−0.2380), highlighting its inability to generalize for this task effectively. The error analysis across different models highlights critical factors influencing prediction performance for the given tasks.

Table 6. Results for Hot fluid outlet temperature (T4) (80–20 Split)

Model	R^2	RMSE	MAE
Linear Regression	0.9336	0.3917	0.2907
Decision Tree	0.5498	0.8760	1.0195
Random Forest	0.3011	1.2703	1.6136
Support Vector Machines	−0.2380	1.4689	1.6906
Gradient Boosting Machines	0.4191	1.1580	0.8613

Linear Regression (LR) consistently achieved near-perfect results ($R^2 \approx 1.0$) with minimal RMSE and MAE across predictions, emphasizing the dataset’s strong linear correlations and the suitability of LR for capturing these relationships. Random Forest (RF) and Gradient Boosting Machines (GBM) delivered competitive performance, particularly for tasks requiring moderate non-linearity handling, such as turbulence effects in heat transfer. However, their slightly higher errors compared to LR indicate potential overfitting to localized patterns, reducing generalization, particularly evident in predictions like the hot fluid outlet temperature (T4). Decision Tree (DT), despite offering interpretable results, exhibited significant overfitting, resulting in elevated error metrics and reduced accuracy. Support Vector Regression (SVR), on the other hand, consistently underperformed, with high RMSE and even negative R^2 values in some cases. This can be attributed to its reliance on complex kernel transformations, which proved unnecessary for the dataset’s inherent linear trends, leading to poor generalization. These findings reaffirm the dominance of LR for datasets with strong linear dependencies, while ensemble methods like RF and GBM remain viable alternatives for scenarios involving moderate non-linearity or noise.



4 Conclusions and Future Scope

The results amply highlight the superior performance of the Linear Regression method in the prediction of forced convection heat transfer rate ($\dot{Q}$), when the relationships between the ‘features’ and the ‘targets’ are nearly linear. The Random Forest technique also performs well in accurately predicting both the Nusselt number (Nu) and heat transfer rate ($\dot{Q}$), demonstrating its capability to model non-linear interactions. In contrast, the techniques of Decision Trees and Support Vector Regression show limitations in handling the complexity of the dataset, with the SVR performing poorly across all metrics.

The above findings underline the importance of model selection in achieving accurate predictions for fluid flow and heat transfer in an experimental study on convection. While linear regression is effective for simpler relationships, ensemble methods like the Random Forest offer a more generic solution for capturing non-linear dependencies, making them suitable for complex applications in thermal engineering.

As can be concluded from the present study, the application of machine learning in a practical heat exchanger configuration promoting forced convection heat transfer opens up promising avenues for innovation and practical benefits. By expanding the dataset (from experimental and/or CFD studies) to include diverse exchanger geometries, thermal boundary conditions, and material properties, deep learning models can achieve higher accuracy and robustness. For instance, these models can optimize the design and operation of heat exchangers, reduce energy losses, and ensure flow and thermal stability in critical processes. Moreover, this approach can accelerate the exploration of using novel materials and configurations for augmenting heat transfer rates, supporting advancements in fields such as renewable energy, microelectronics cooling, and aerospace engineering.

As computational power increases and access to high-fidelity experimental and/or simulation data improves, these models have the potential to bridge the gap between

empirical methods and theoretical frameworks, promoting data-driven thermal management solutions. By leveraging these innovations, industries can achieve sustainable and cost-effective solutions for state-of-the-art thermal engineering challenges.

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